

Prof. László T. Kóczy

Subject: A learnable Kóczy fuzzy signature as a differentiable hypergraph layer — early results and a request for your feedback.

Dear Professor Kóczy,

I have been working on an explicit, learnable realisation of fuzzy signatures as a differentiable neural-network layer, and I would like to share the early results with you. The motivation is your classical construction: a fuzzy signature is a hierarchical tree of fuzzy concepts in which internal nodes are aggregation operators (t-norms, t-conorms, OWA) and leaves carry elementary memberships. Almost everything in the construction is interpretable; only the membership functions and the aggregator weights are tuned in the classical setting.

The construction I have built keeps the structure but makes *every* learnable element a named fuzzy-systems primitive:

- Each per-vertex feature is a learnable Catmull–Rom membership function $\mu_\theta: X \rightarrow [0, 1]$.
- The signed branches of the architecture are an Atanassov intuitionistic-fuzzy pair (μ^+, μ^-) with hesitancy $\pi = 1 - \mu^+ - \mu^-$.
- The polarity gate that mixes the pair is a Zadeh sigmoidal hedge with learnable dilation τ per channel.
- Vertex-to-edge aggregation is a categorical t-norm ($T_G / T_P / T_L$); edge-to-vertex is a categorical t-conorm.
- Multi-arity (multi-scale) fusion across receptive-field sizes is an OWA with α_k softmax weights.
- The output head is a Sugeno weighted-average defuzzification followed by the TSK output rule.

I attach a 20-page mathematical background that lays out this correspondence in full, along with the architectural plan and a robotic-collaboration application story.

Empirically, the architecture has been validated on MNIST at a small parameter budget (test_acc 0.5354 at 9,642 parameters; the α_k multi-scale mixer emergently specialises so that the first layer prefers the fine receptive field and the last layer the coarse one — exactly the multi-scale story HSiKAN-vision relies on, re-emerging in the fuzzy-signature setting). The architecture also extends to a pose-detection variant; the validated single-frame synthetic-pose model is now the basis for a planned two-robot collaborative-behavior test bed.

I emphasise the honest current limits: the MNIST result is single-seed; the architecture is not yet at HSiKAN parity ($\sim 0.95+$); the pose-detection comparison against simpler baselines is in flight. The point of this letter is to ask for your feedback on the direction, not to claim a final result. In particular:

- Does the named-primitive correspondence preserve what you consider essential about fuzzy signatures?
- Is the Atanassov-pair reading of the signed-branch construction the right inductive bias for an *intuitionistic* fuzzy signature, or do you see a different formulation that better captures hesitancy?

- For the robotic-collaboration story (two-robot test bed, cooperative-handover / competitive-grasp / independent-parallel interactions), does the framework's hierarchical aggregation align with how you envision fuzzy signatures supporting human-robot collaboration?

I would value any feedback you can offer, and I am happy to share the full implementation, the test matrix, or any other technical detail you would like to see.

With deep respect for your foundational work,

[SENDER]

Attachments:

1. `background.pdf` — Mathematical Background for the Learnable Kóczy Fuzzy Signature Layer (20 pp). Atanassov IFS, classical t-norms / t-conorms / negations, Kóczy fuzzy signatures, defuzzification methods, Zadeh linguistic hedges, OWA aggregators, the full HSiKAN-Kóczy correspondence table, $[0, 1]$ -preservation theorem, TSK-output-rule derivation, worked examples.
2. `plan_fuzzy_signature.pdf` — Architectural plan for the FuzzySignatureLayer (7 pp).
3. `plan_robotics.pdf` — Robotics / behavior / collaboration application plan with the two-robot test case (7 pp).